

Digital Signal Processing 2

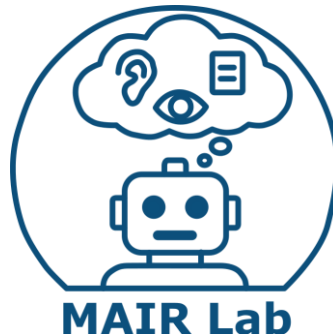
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Speech And Audio Recognition
(오디오 음성인식)

<https://mairlab.kookmin.ac.kr/>



This lecture material refers to
https://github.com/yandexdataschool/speech_course?tab=readme-ov-file and
<https://github.com/markovka17/dla>



Discrete Fourier Transform (DFT)

- Discrete Fourier Transform (DFT):

$$X[k] = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{k}{N}n}$$

- Inverse Discrete Fourier Transform (IDFT):

$$x[n] = \sum_{k=0}^{N-1} X[k] e^{j2\pi \frac{n}{N}k}$$

Discrete Fourier Transform (DFT)

- $X = M \cdot x$, where $x = \{x[0], \dots, x[N-1]\}$ and $X = \{X[0], \dots, X[N-1]\}$

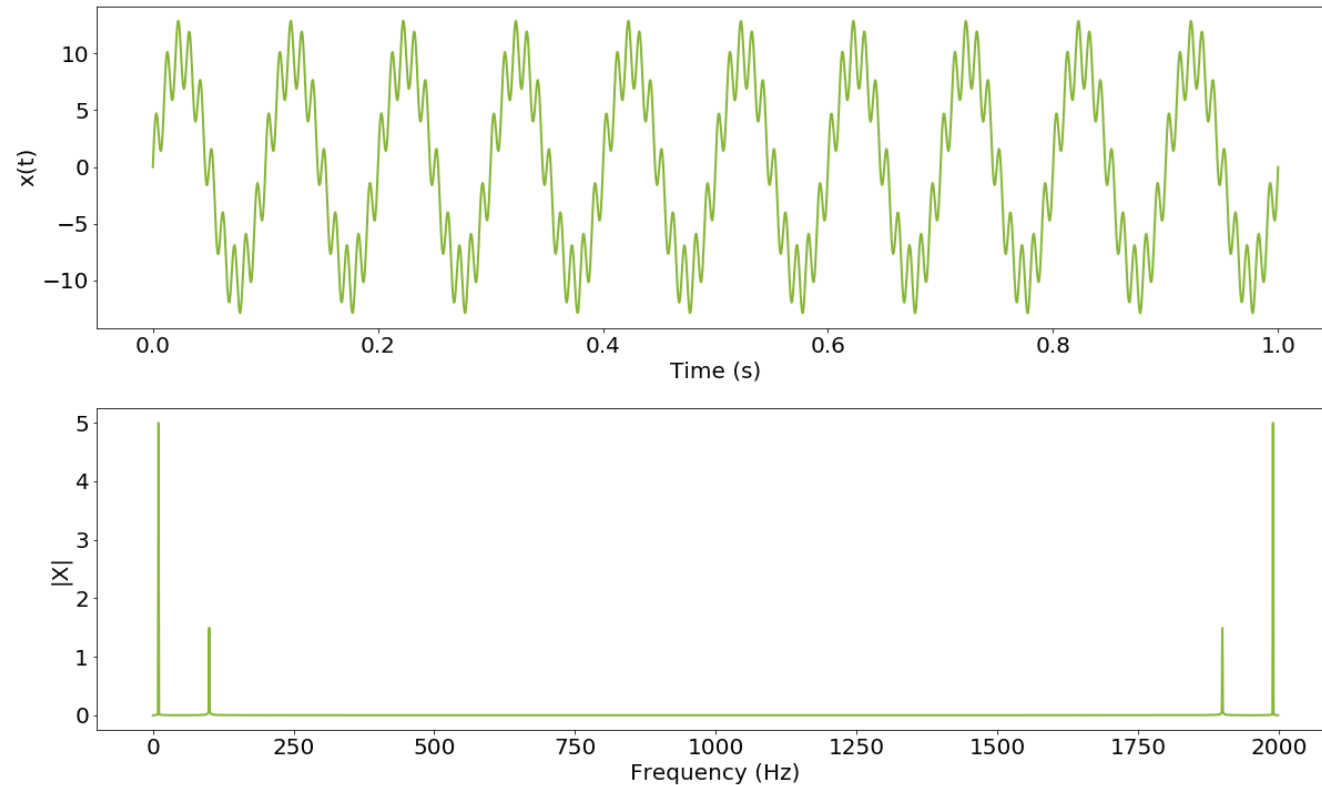
$$M_{mn} = \exp\left(-2\pi i \frac{(m-1)(n-1)}{N}\right)$$

$$\mathbf{M} = \begin{pmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 1 & e^{-\frac{2\pi i}{N}} & e^{-\frac{4\pi i}{N}} & e^{-\frac{6\pi i}{N}} & \dots & e^{-\frac{2\pi i}{N}(N-1)} \\ 1 & e^{-\frac{4\pi i}{N}} & e^{-\frac{8\pi i}{N}} & e^{-\frac{12\pi i}{N}} & \dots & e^{-\frac{2\pi i}{N}2(N-1)} \\ 1 & e^{-\frac{6\pi i}{N}} & e^{-\frac{12\pi i}{N}} & e^{-\frac{18\pi i}{N}} & \dots & e^{-\frac{2\pi i}{N}3(N-1)} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & e^{-\frac{2\pi i}{N}(N-1)} & e^{-\frac{2\pi i}{N}2(N-1)} & e^{-\frac{2\pi i}{N}3(N-1)} & \dots & e^{-\frac{2\pi i}{N}(N-1)^2} \end{pmatrix}$$

We can compute it with FFT, which is extremely fast (theoretically $O(N \log N)$ for signal of size N)

Discrete Fourier Transform (DFT)

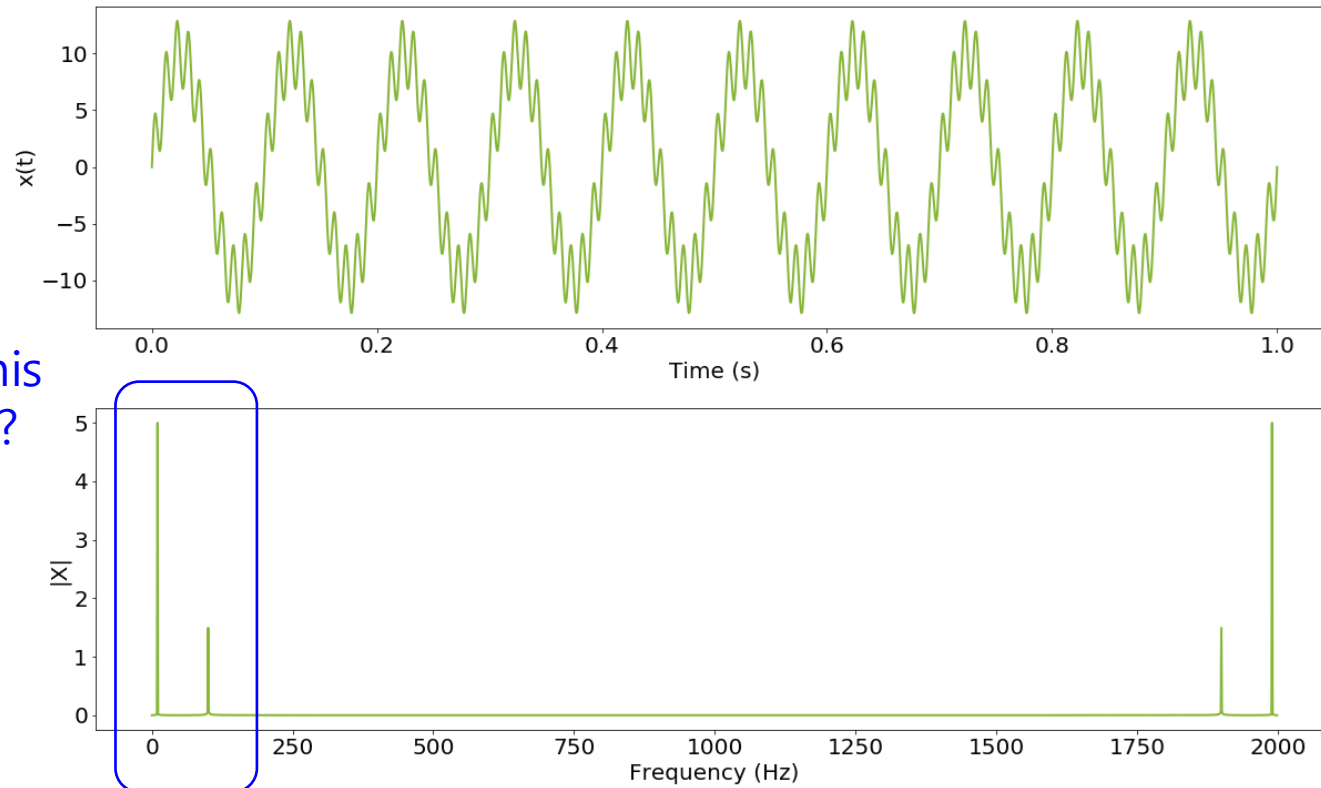
- Example of DFT:
 - $f(t) = 10 \sin(2\pi 10t) + 3 \sin(2\pi 100t)$



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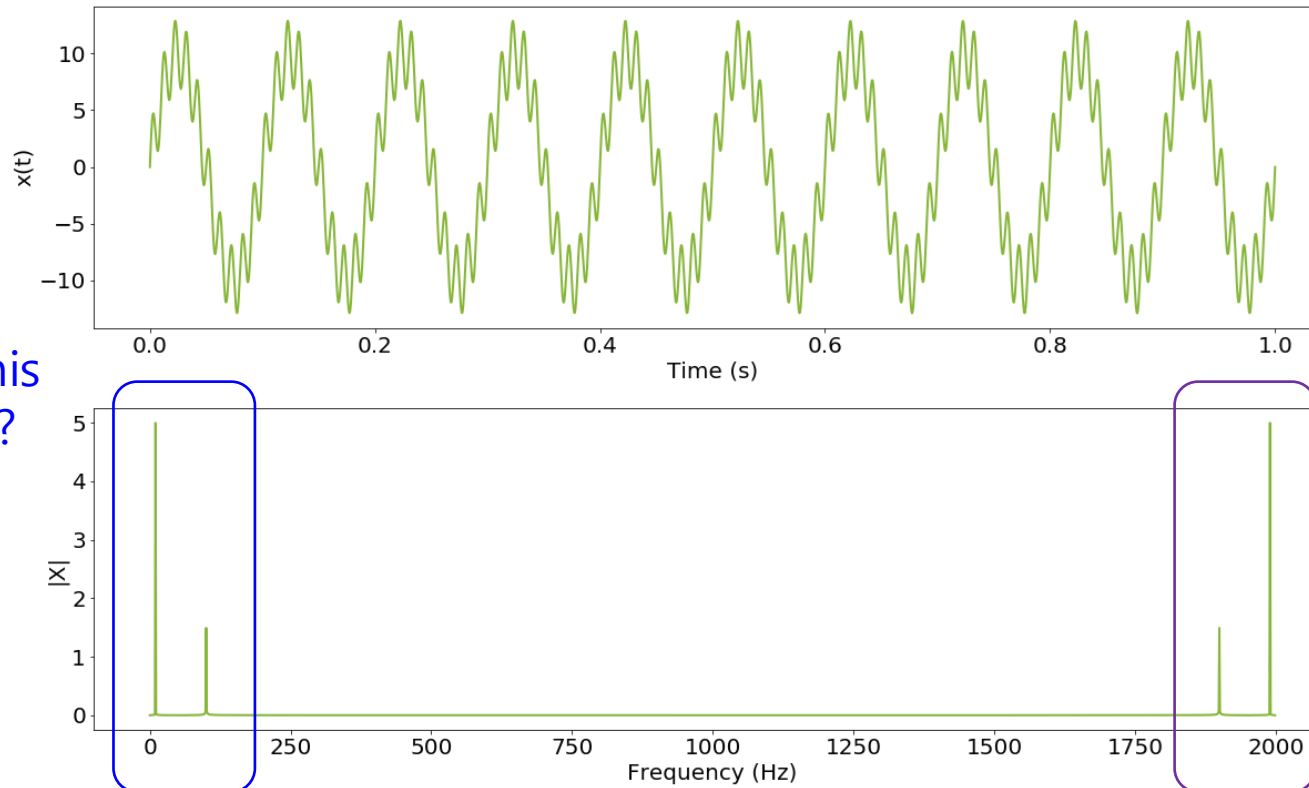
1. What does this graph indicate?



Discrete Fourier Transform (DFT)

- Example of DFT:
 - $f(t) = 10 \sin(2\pi 10t) + 3 \sin(2\pi 100t)$

1. What does this graph indicate?



2. Why does this part keep repeating?

Discrete Fourier Transform (DFT)

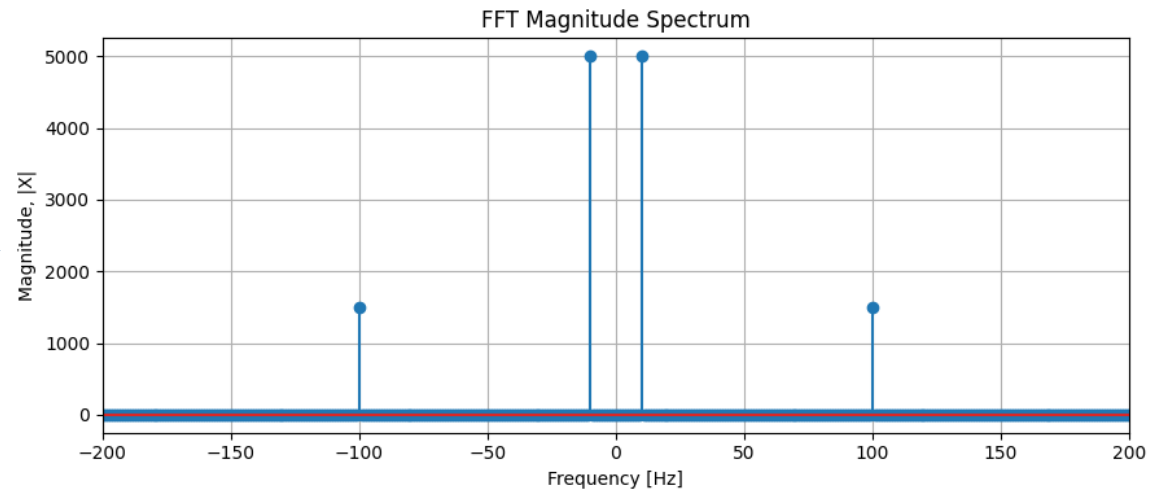
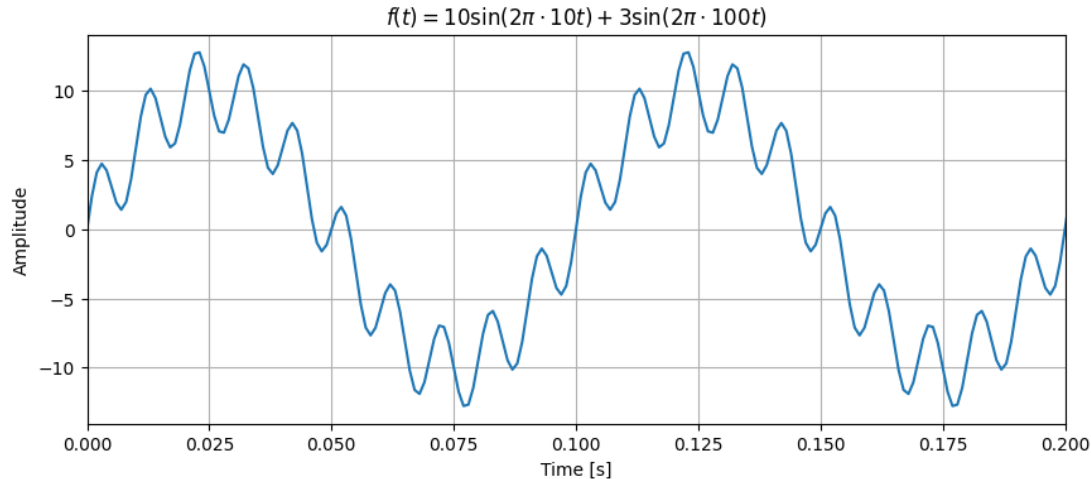
- Example of DFT:
 - $f(t) = 10 \sin(2\pi 10t) + 3 \sin(2\pi 100t)$

$$\begin{aligned} X_m &= \sum_{n=0}^{N-1} x_n \exp\left(-j2\pi \frac{m}{N} n\right) \\ X_{N-m} &= \sum_{n=0}^{N-1} x_n \exp\left(-j2\pi \frac{N-m}{N} n\right) \\ &= \sum_{n=0}^{N-1} x_n \exp\left(-j2\pi n + j2\pi \frac{m}{N} n\right) \\ &= \sum_{n=0}^{N-1} x_n \exp\left(j2\pi \frac{m}{N} n\right) \\ &= (X_m)^* \end{aligned}$$

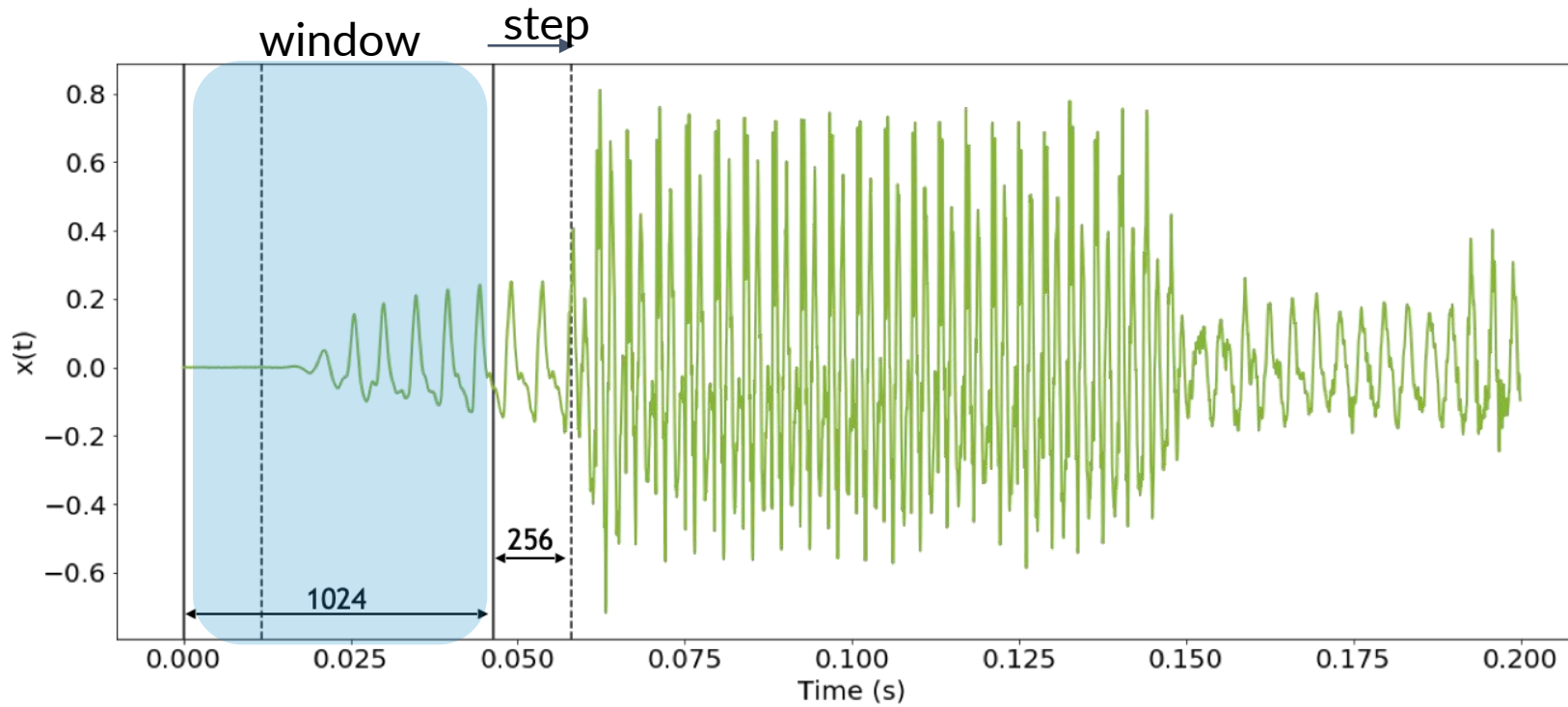
$$\begin{aligned} X_m &= a_k(\cos(\phi_k) - i \sin(\phi_k)) \\ (X_m)^* &= a_k(\cos(\phi_k) + i \sin(\phi_k)) \end{aligned}$$

Practice: Discrete Fourier Transform (DFT)

- Colab Practice!
- Example of DFT:
 - $f(t) = 10 \sin(2\pi 10t) + 3 \sin(2\pi 100t)$

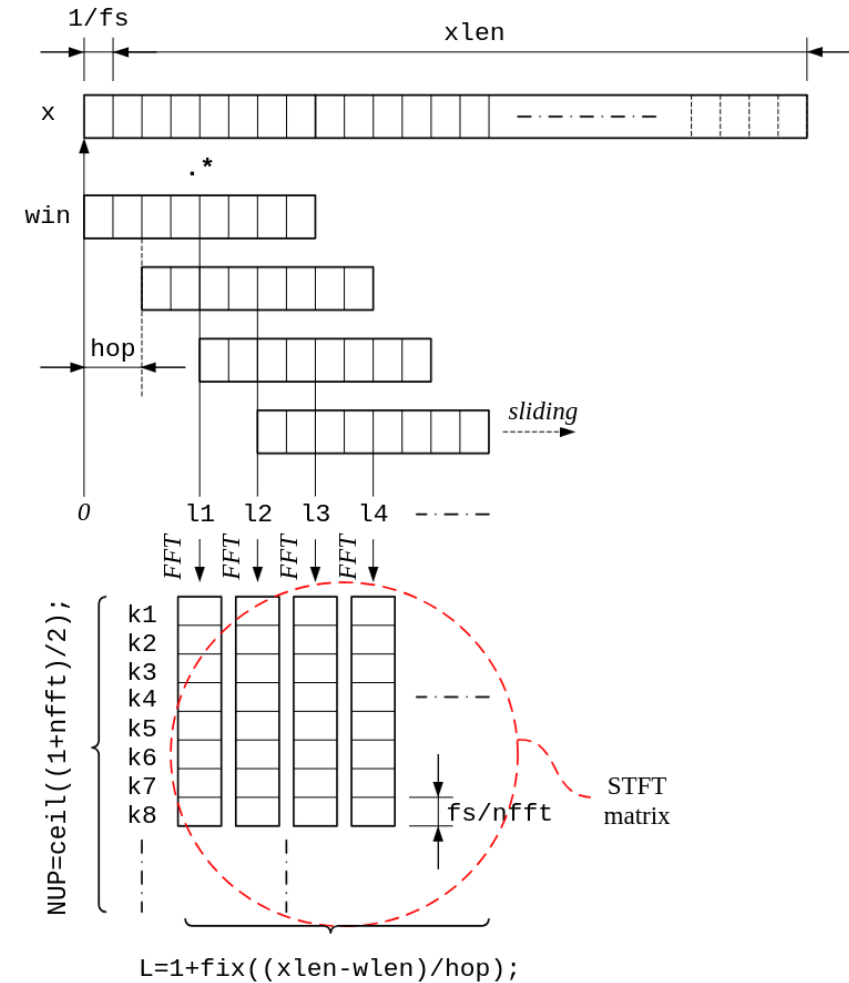
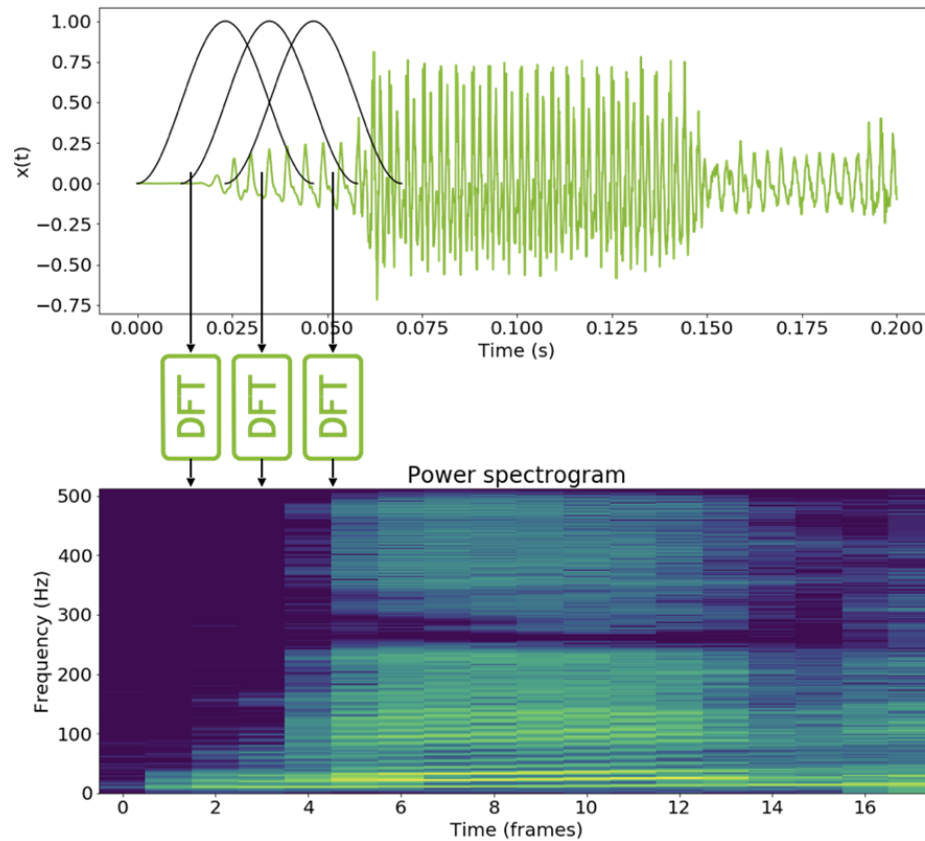


Shor-Time Fourier Transform



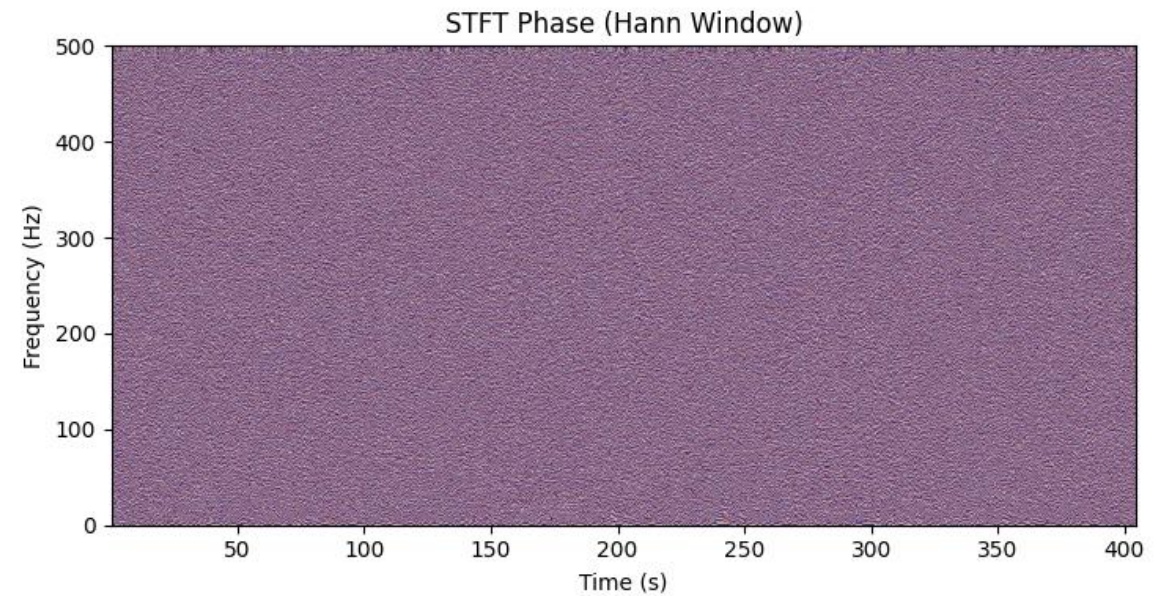
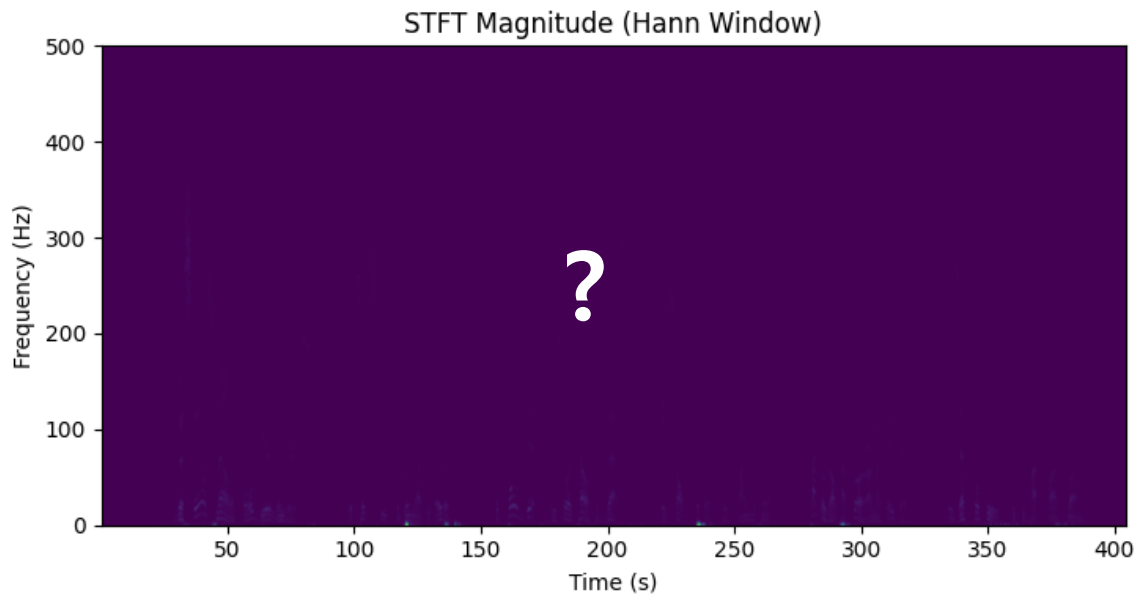
Shor-Time Fourier Transform

- STFT + Window function



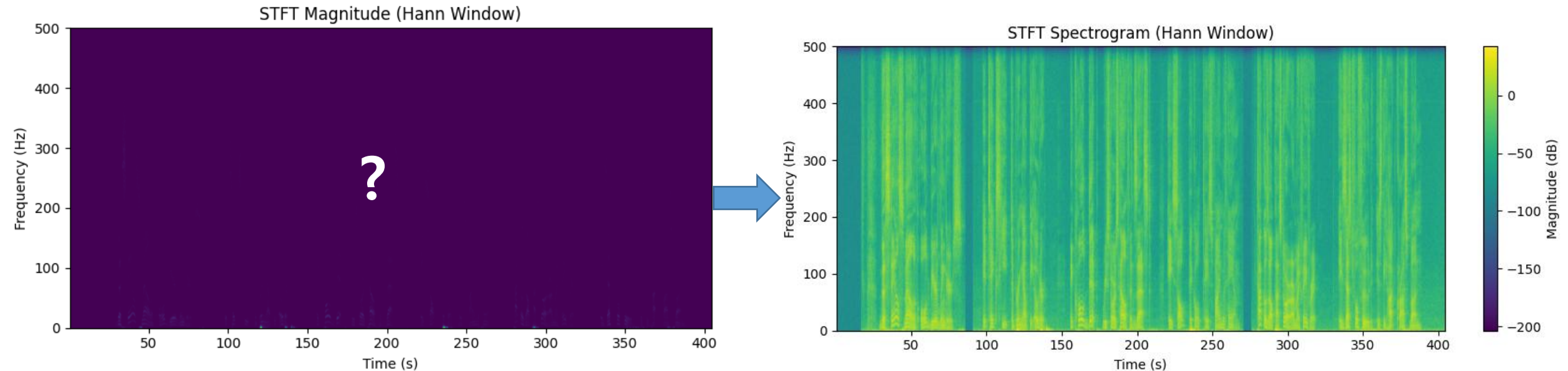
Shor-Time Fourier Transform

- Spectrogram:
 - The output of STFT is a complex matrix of size $n\text{Freq} * n\text{Windows}$
 - Useually, we look at magnitude and phase of the output



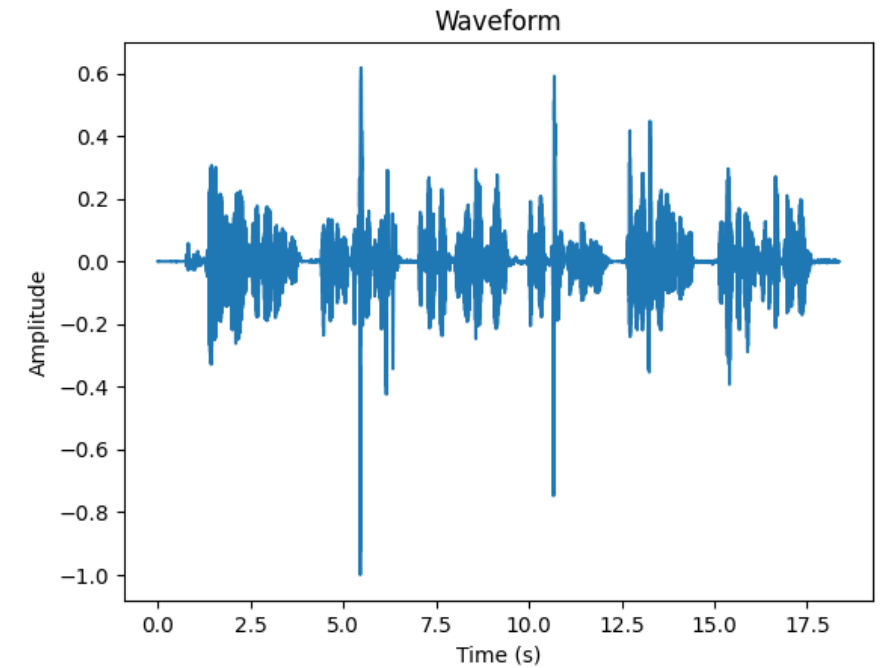
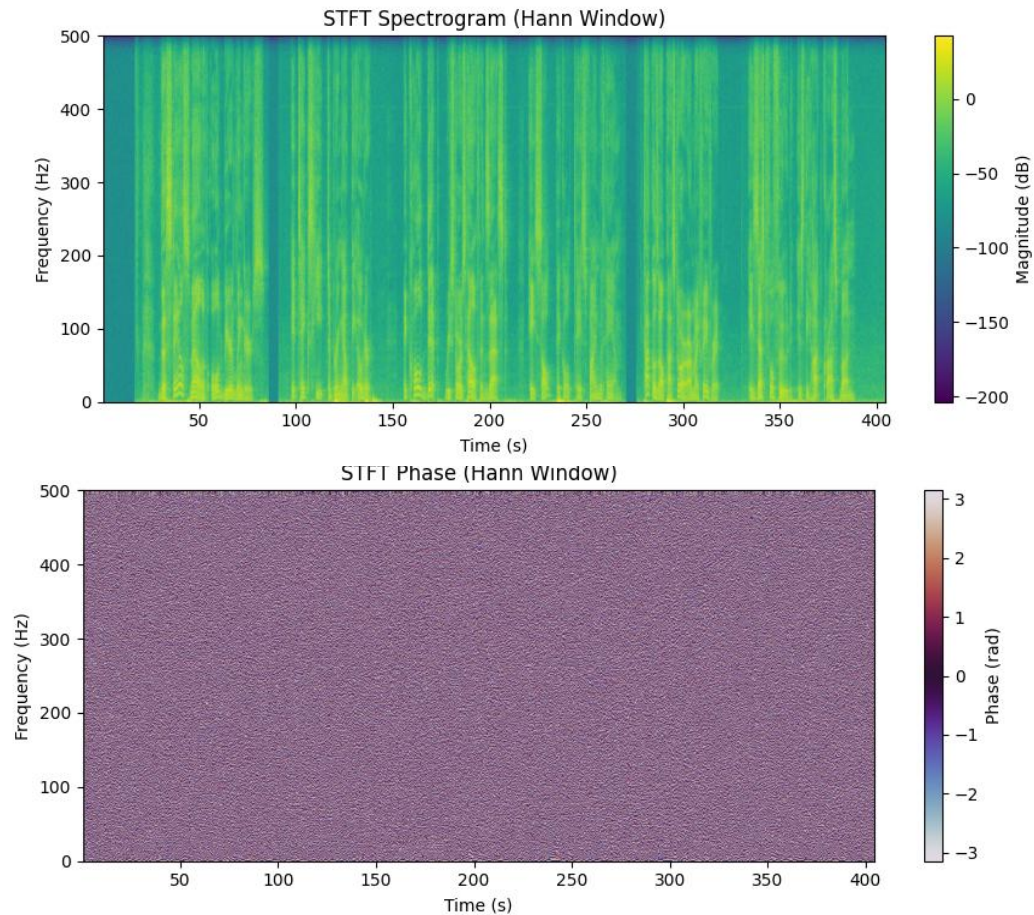
Shor-Time Fourier Transform

- Spectrogram:
 - The logarithm of the magnitude spectrogram is much easier visually to interpret



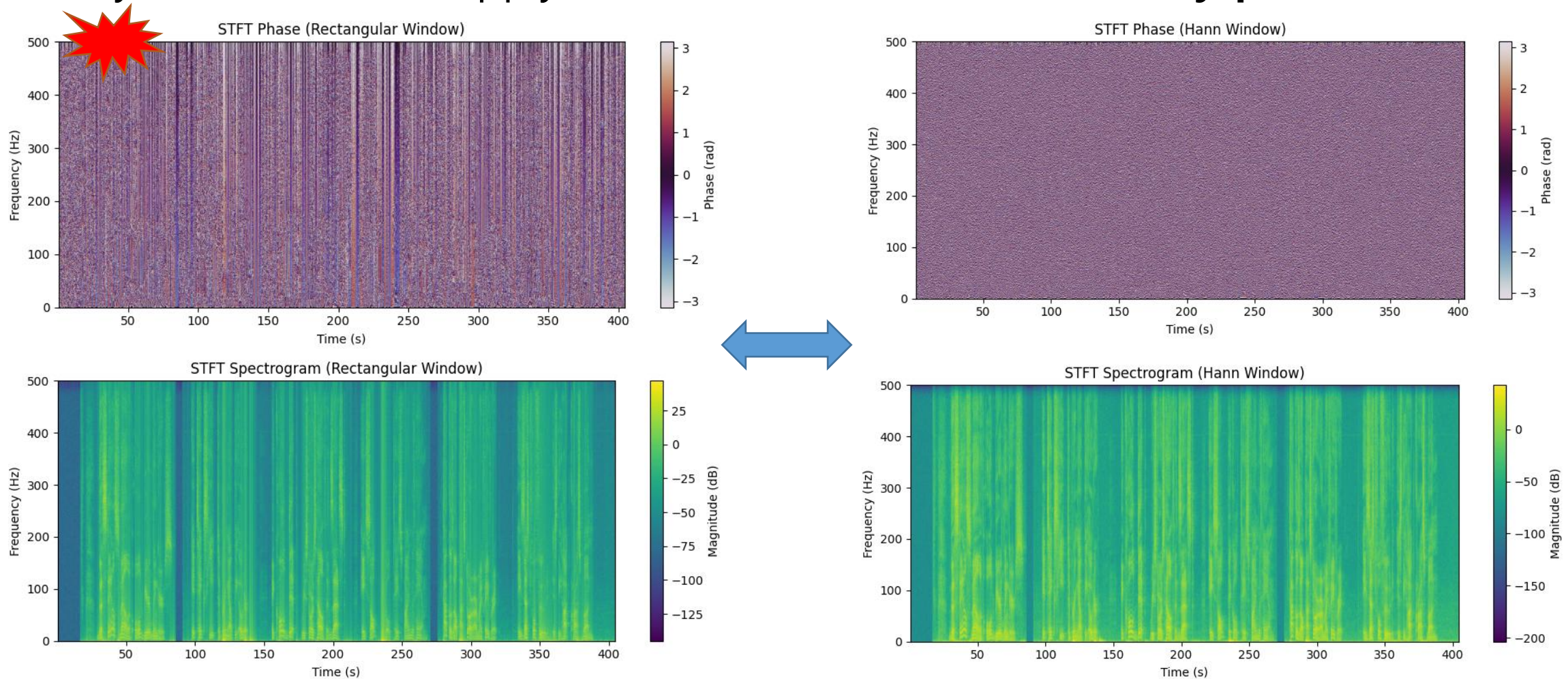
Shor-Time Fourier Transform

- Inver STFT (iSTFT)
 - STFT result can be inverted back given the parameters are known (window, hop and step sizes)



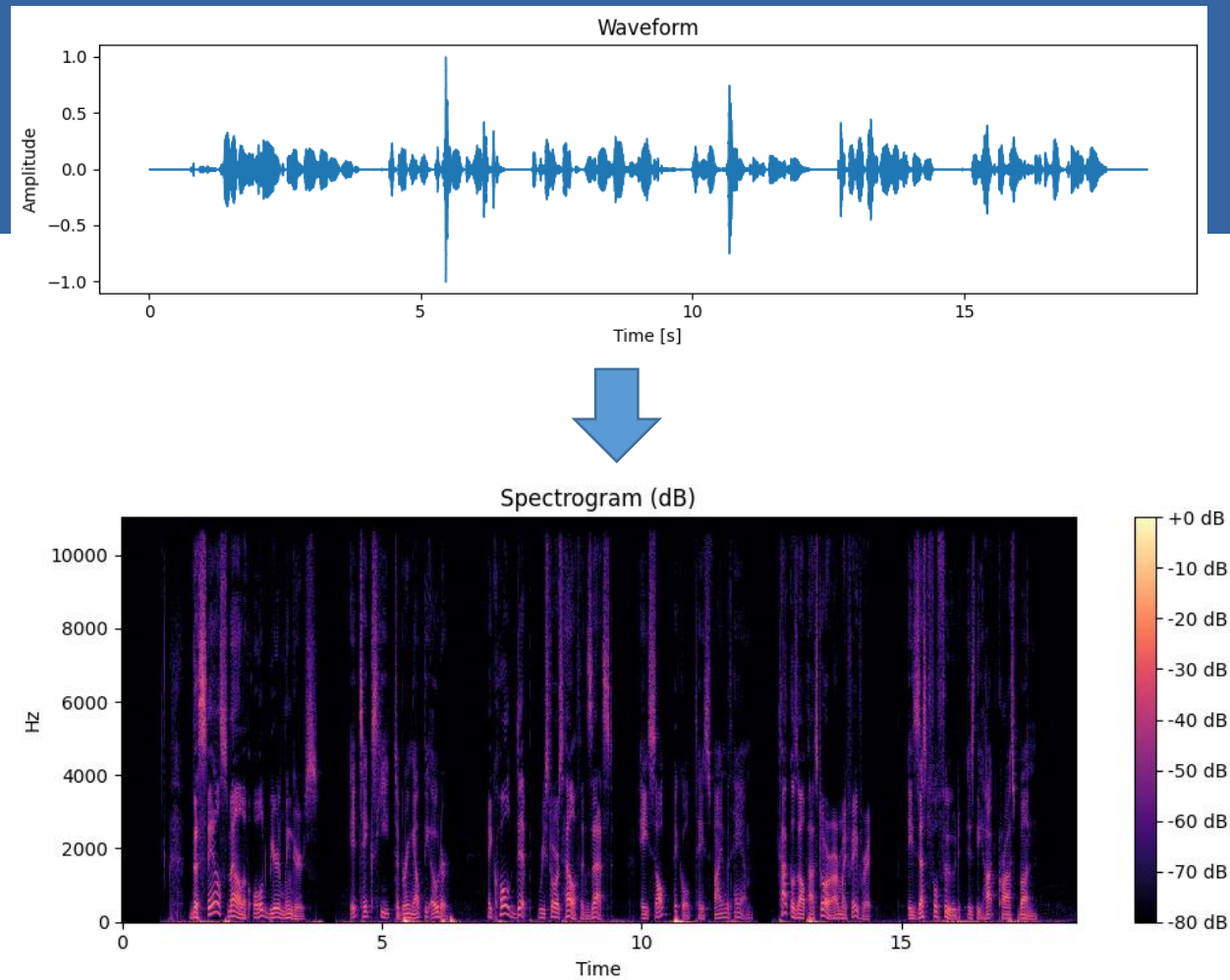
Shor-Time Fourier Transform

- Why do we have to apply window functions: **discontinuity problem!**



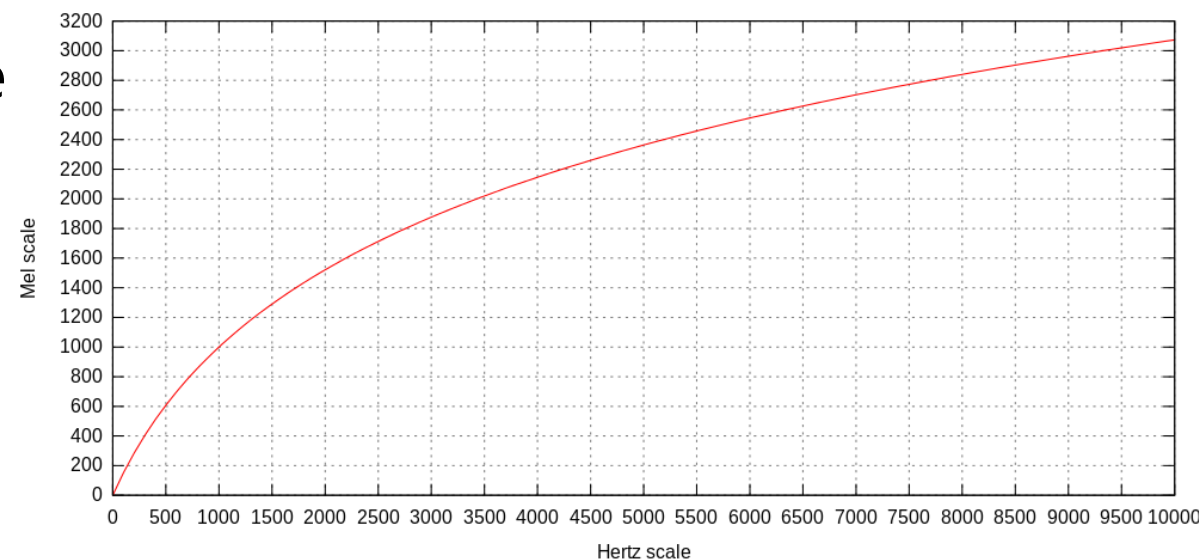
Practice!

- Colab practice!



Mel Spectrogram

- Humans perceive sound on a log-scale
- For human ear:
 - 500 Hz << 600 Hz
 - but 5000 Hz ≈ 5100 Hz



There is no single mel-scale formula.^[3] The popular formula from O'Shaughnessy's book can be expressed with different logarithmic bases:

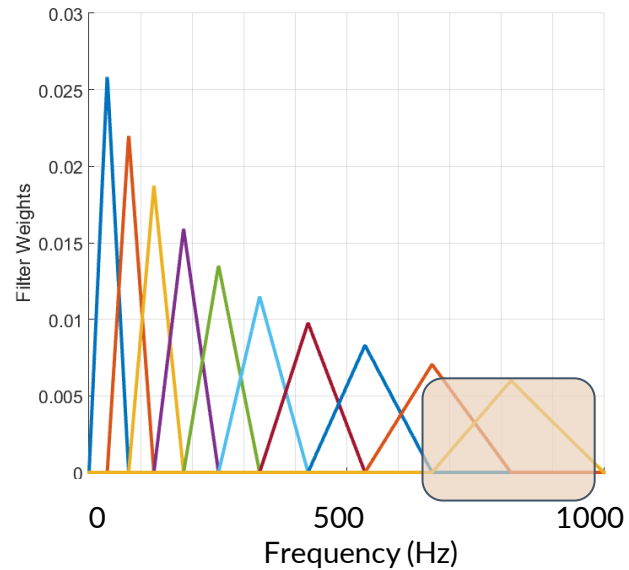
$$m = 2595 \log_{10} \left(1 + \frac{f}{700} \right) = 1127 \ln \left(1 + \frac{f}{700} \right)$$

The corresponding inverse expressions are:

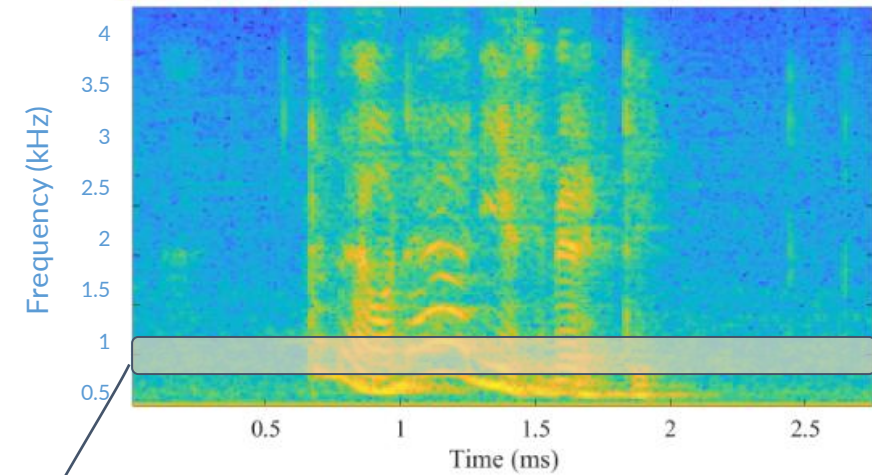
$$f = 700 \left(10^{\frac{m}{2595}} - 1 \right) = 700 \left(e^{\frac{m}{1127}} - 1 \right)$$

Mel Spectrogram

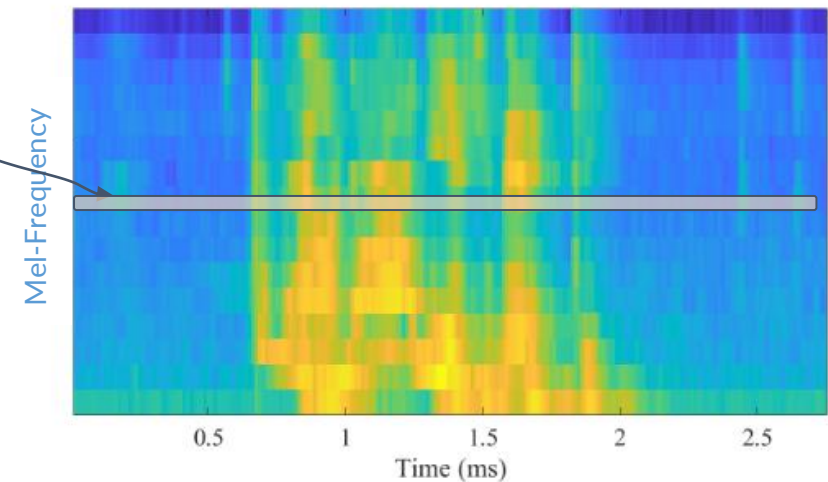
- Mel Spectrogram



Spectrogram of a segment of speech

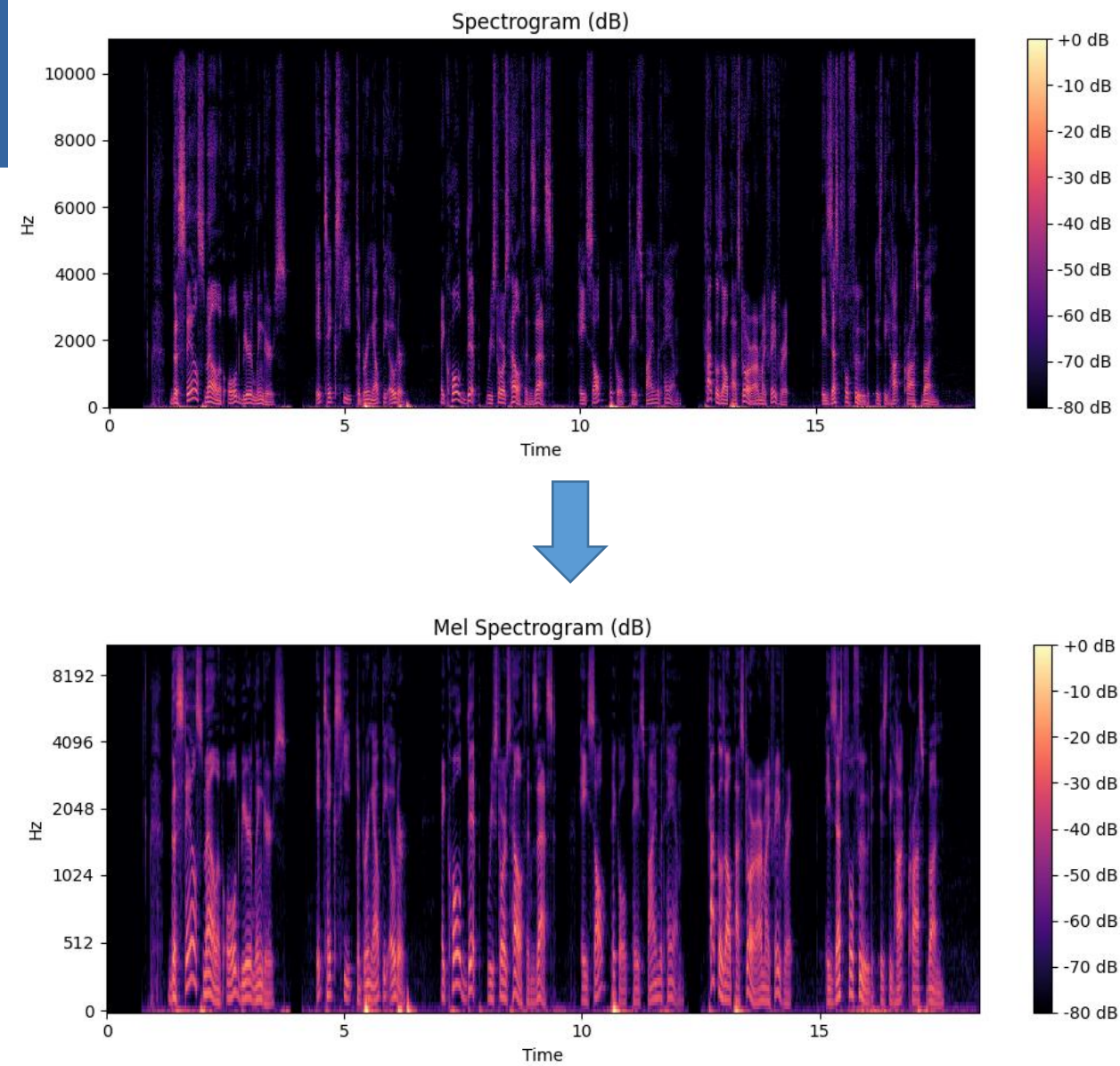


Spectrogram after multiplication with mel-weighted filterbank



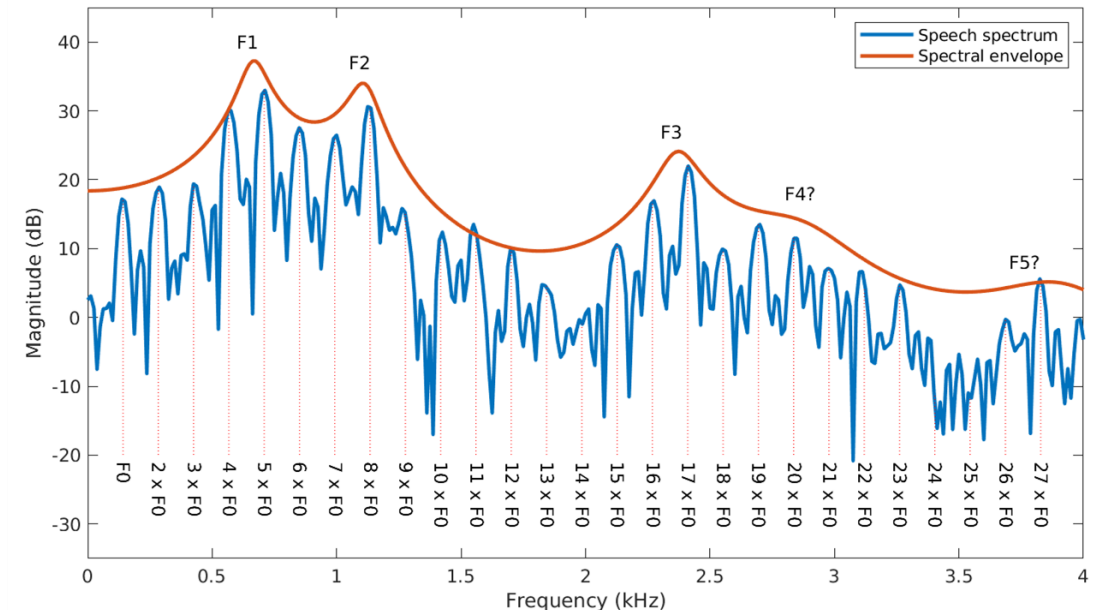
Practice!

- Colab practice!



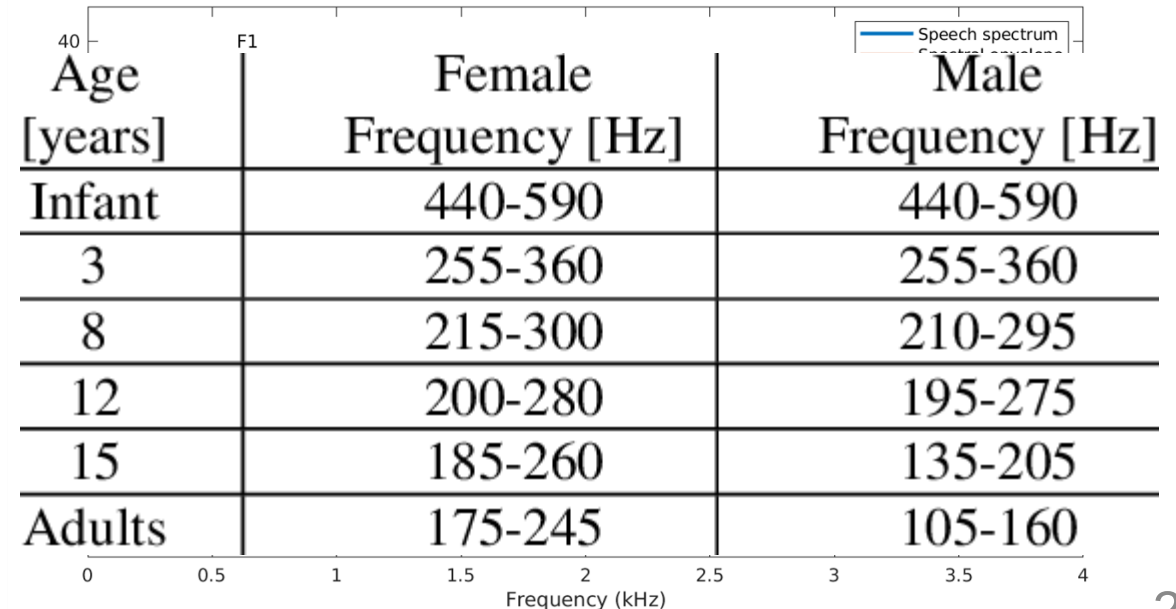
MFCC

- Fundamental frequency is the physical source frequency, but there are **resonances** (공진) and **harmonics** (배음)
- Peaks on envelope curve are **formants**
- Pitch is perceptual value, **F0 is physical, harmonics are $k \times F0$**
- For speech F0 lie roughly in the range 80 to 450 Hz, typically males have lower voices than females and children



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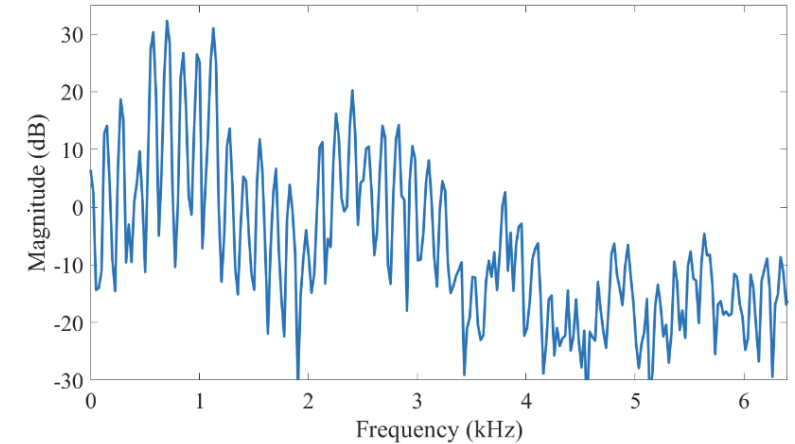
MFCC

- Cepstrum

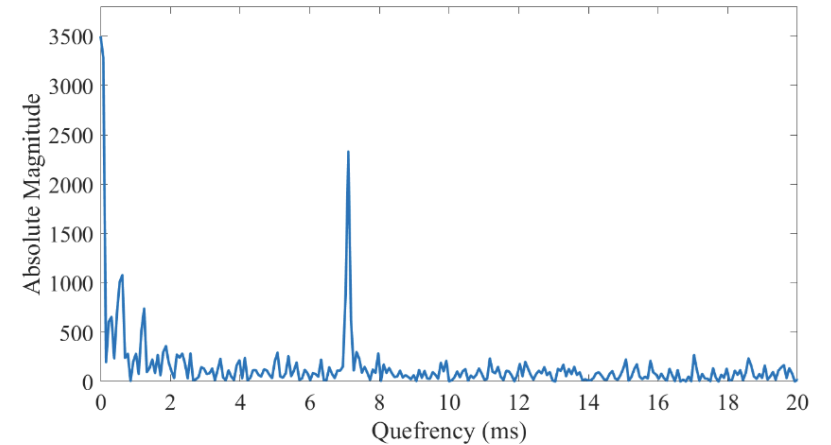
- Fourier spectrum of voice has periodic structure
- Apply Inverse DFT to log-spectrum ($\log|X(\omega)|$) and obtain Cepstrum
- Peak in Cepstrum should be located at $\frac{1}{F_0}$

왜 이럴까?

Log-spectrum of speech segment



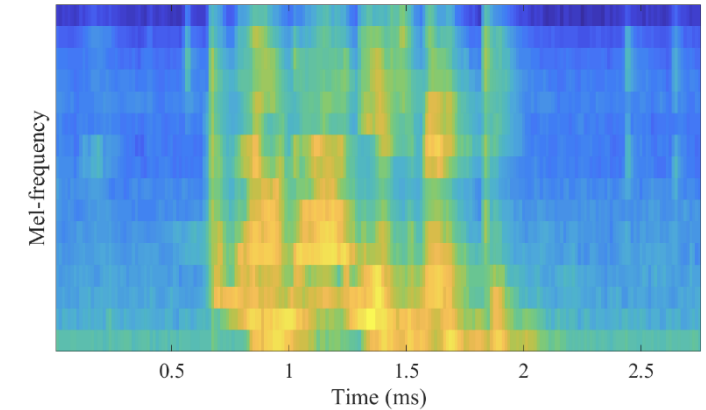
Cepstrum of speech segment



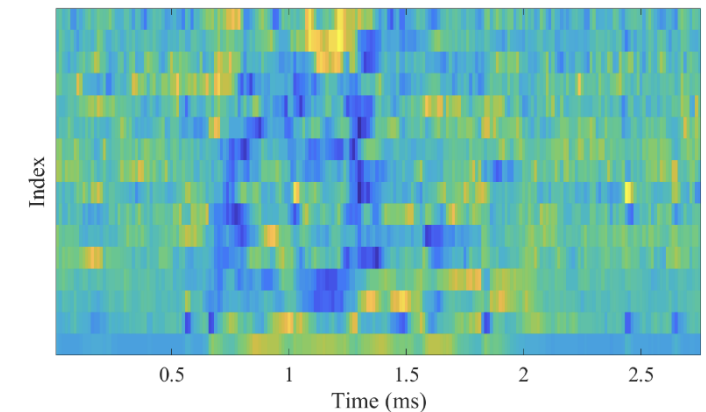
MFCC

- Mel-Frequency Cepstral Coefficient (MFCC)
 - Apply STFT to the signal
 - Apply mel filters
 - Take the log value
 - Apply Discrete Cosine Transform

Spectrogram after multiplication with mel-weighted filterbank



Corresponding MFCCs



Practice!

- Colab practice!

